

SUJIT SHRESTHA

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I am a passionate and forward-thinking Mechanical Engineer currently pursuing a Master of Professional Engineering at Monash University. My experience includes leading a startup, innovating in product design, healthcare, and conducting material science research. With strong skills in CAD modeling, I excel at designing and developing precise, efficient, and practical solutions. I am deeply committed to advancing the fields of energy, sustainability, and manufacturing, striving to create impactful and sustainable technologies.

EDUCATION

Master's of Professional Engineering

July 2024-Present

Monash University

Mechanical Engineering

Bachelor of Engineering

June 2018- December 2023

Kathmandu University

Major in Mechanical Engineering

CGPA : 3.27

PROFESSIONAL EXPERIENCE

Product Member

January 2024-Present

Precious Plastic Monash

A student-run circular design initiative at Monash University that transforms waste plastics into useful products using digital manufacturing.

- Designing and prototyping recycled-plastic products.
- CAD modelling and preparing files for manufacturing.
- Operating machines (Water-jet cutter, Laser cutter, Shredder, Sheet press, 3D printers).
- Testing tolerances, assembly fit, and material behaviour.
- Supporting circular economy initiatives on campus.
- Completed routine tasks and participated in team meetings.

Project Intern

June 2025-July 2025

Allnex & Monash University

Global virtual internship with Monash and Warwick students, completing a project for chemical company Allnex.

- Collaborated with Monash (Australia & Malaysia) and Warwick students in a global virtual internship.
- Designed a CSR initiative for Allnex to improve employee engagement and social impact.

- Developed a school-focused CSR program in Malaysia with timelines, benefits, and stakeholder insights.
- Contributed research, problem-solving, and cross-cultural teamwork to provide recommendations for Allnex.
- Presented project outcomes and recommendations to Allnex as part of the Monash–Warwick Alliance.

Co-founder and Engineering lead

November 2022-July 2024

Paramendo Nepal

A recycling initiative that transforms plastic waste into useful products while promoting circular economy practices in rural Nepal.

- Led the team as CEO, overseeing strategic planning, decision-making, and overall company direction.
- Managed the day-to-day operations of the organization, ensuring smooth workflow and efficiency.
- Directed product development and prototyping focused on recycling plastics, creating sustainable solutions for waste management.
- Conducted research on material properties, advancing the performance and durability of recycled plastic products.
- Facilitated community engagement in villages of Nepal, raising awareness about waste management practices and promoting sustainable solutions.

Engineering Intern

February 2023-May 2023

Impact Hub Kathmandu

A collaboration between Impact Hub Kathmandu, the Spinal Injury Rehabilitation Center (SIRC), and Rehab Lab France to develop assistive and rehabilitation-focused solutions.

- Development of Occupational and psychotherapy devices using digital manufacturing for spinal injury patients.
- Operation and Maintenance of 3D printer (Ender 3 pro).
- Digital manufacturing and design thinking training of medical professionals at SIRC.
- Passive prosthetic leg development for amputated cats.
- Work featured in a publication by Alessandra Campoli (Open University, UK) in the International Journal of Co-Creation in Design and the Arts.

Research Intern

March 2021-November 2022

Energy System and Technology Research Laboratory (ESTRL), Kathmandu University

- Conducted research on battery recycling in Nepal, focusing on sustainable waste management and resource recovery.
- Presented a research paper at University Scholar, showcasing innovative insights into energy and sustainability.
- Explored the implementation of the AHP model for decision-making in electric vehicle adoption and policy frameworks.

- Organized and managed workshops on battery technologies, solar heaters, clean cooking solutions, and thermal management, fostering knowledge dissemination and practical applications.

TECHNOLOGY SKILLS

- **Design & CAD:** SolidWorks, Fusion 360, AutoCAD, Creo
- **Simulation & Analysis:** ANSYS, Transys, Metronome
- **Digital Manufacturing:** 3D printing (FDM, LPBF basics), Laser cutting, Water-jet cutting, CNC
- **Manufacturing:** Soldering, Routing, Welding, Sand Casting, Image scanning
- **Software & Tools:** MS Excel, MS Word, MS PowerPoint, LabVIEW
- **Programming:** C, C++ , Arduino programming, MATLAB
- **Recycling & Materials:** Plastic sheet pressing, Shredding, Material testing
- **Robotics:** KUKA robot basics
- **Other:** Photography, Videography, Video editing

VOLUNTARY EXPERIENCE

Volunteer – Community Engagement (Precious Plastic Monash)

July, 2025

Family Fiesta Event

- Represented Precious Plastic Monash at the Family Fiesta community event.
- Demonstrated plastic recycling by helping children inject carabiners using recycled plastic.
- Educated families and young participants about recycling, sustainability, and circular design.
- Showcased products made from recycled plastics and promoted PPM's mission to the public.

Digital Manufacturing trainer for Medical Personnel

February 2023 – May 2023

Spinal Injury Rehabilitation Center (SIRC), Nepal

- Delivered digital manufacturing training to healthcare staff, focusing on 3D printing, CAD, human-centred design, and assistive device development.
- Trained Occupational Therapy and Prosthetics & Orthotics departments in using 3D printing for therapy tools and prosthetic applications.
- Conducted hands-on sessions, supported patient therapy activities, and evaluated training effectiveness through case studies.
- Demonstrated strong professionalism, reliability, and enthusiasm, consistently seeking opportunities to learn and contribute.

Instructor - SolidWorks & ANSYS Training

July 2022

SEDS Kathmandu University

- Delivered a 5-day training program for students from 1st to 4th year on SolidWorks and ANSYS fundamentals.

- Prepared structured presentations and hands-on sessions covering modelling, simulation, and design workflows.
- Guided students through practical exercises, improving their CAD and FEA skills.

HOBBIES

Soccer, Hiking, Photography, Birdwatching, Guitar, Sustainable product design

REFEREES

Available upon request.